

JOSHWYN PAREKH

Computer Engineer @ University of Waterloo | Expected Graduation: Apr. 2027
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</> SKILLS

Languages: C, C++, C#, Python, TypeScript, SQL, JavaScript, Verilog, VHDL, Kotlin

Frameworks & Cloud: Node.js, React, .NET, Pandas, Scikit-learn, AWS, GCP, Kubernetes, Docker

Hardware & Tools: Altium Designer, LTspice, MATLAB, PostgreSQL, MongoDB, Git, Linux, Cursor, Claude, UE5, Unity

Interpersonal skills: Lvl-1 tennis and ski coach, strengthened critical thinking, collaboration, and communication skills

⚙️ EXPERIENCE

Full Stack Software Developer

Jan. 2026 – Apr. 2026

Geotab

Waterloo

- Developed full-stack data pipelines for the Tallon team to ensure California Air Resources Board (CARB) compliance, processing real-time commercial fleet telematics using C#, .NET, and PostgreSQL.
- Refactored the Driver Vehicle Inspection Report (DVIR) interface into tablet-responsive layouts based on Figma specifications, completing end-to-end development utilizing TypeScript, React, and Node.js.
- Optimized backend queries and front-end state management across the inspections platform deployed on Google Cloud Platform (GCP), improving system reliability and reducing manual processing errors.

Power Team Lead

Sep. 2025 – Present

WATurbine Design Team

Waterloo

- Led the design and implementation of an MPPT charge controller and custom circuit boards, integrating power systems for a small-scale wind turbine.
- Architected a fault-tolerant power distribution network, integrating real-time sensor feedback to dynamically manage variable wind speeds and mitigate loss-of-load and battery overcharge conditions.

Virtual Reality Game Developer

May 2025 – Aug. 2025

Liftwerx

Cambridge

- Designed and developed interactive VR training experiences in Unity using C#, applying game development principles to create immersive, gamified work procedures for the LiftWerx VR Academy training program.
- Built and refined 3D game assets and environments using Blender, SolidWorks, and Unity, supporting fast-paced prototyping and production of polished, gameplay-driven VR content for on-site technicians.

Systems Software Developer

Sep. 2024 – Dec. 2024

Co-operators

Guelph

- Led the end-to-end development of a company-wide automated notification system, independently managing all stages of the software development life cycle from design and implementation to deployment and maintenance.
- Integrated multiple enterprise applications using APIs, secure authentication and web services to enable seamless communication for data collection within the automated systems.
- Built an AWS-based chatbot using the ChatGPT API, HTTP requests, and Lambda functions to query a database to be able to dynamically retrieve and validate data from a database to deliver accurate responses.

⚙️ PROJECTS - CHECK OUT MY [PORTFOLIO!](#)

LLM Hardware Block Implemented on an FPGA | *SystemVerilog, Pynq Board*

Jan. 2025 – Apr. 2025

- Implemented the BERT transformer model (a core architecture behind modern Large Language Models) using integer-only operations which learns word relationships and context.
- Built all key modules (quantization, softmax, normalization, GELU) in SystemVerilog at the RTL level, and deployed the model on a Pynq FPGA board achieving highly efficient hardware execution.

Advanced Multi-threaded Image Retrieval and Assembly | *C, cURL, pthreads*

May 2024 – Jun. 2024

- Developed a multi-threaded application to retrieve and assemble image segments concurrently from web servers. Used the pthreads library to reduce latency by efficiently distributing the workload across multiple servers.

“ TESTIMONIALS

The following testimonials highlight my performance across various co-op positions I've completed during my degree at the University of Waterloo. Each testimonial is from my direct manager, all of whom rated my work terms as **OUTSTANDING**.

Andre Larocque

LiftWerx

Rating: Outstanding

“Joshwyn exceeded expectations in terms of the quality of his work, the quantity of valuable output, and his contributions to projects beyond the scope of his role. . . You are a capable, talented, intelligent developer and designer. Your critical insights and the reliability of your word are invaluable to any workplace.”

- Recognized as a talented, reliable developer and collaborator.
- Exceeded expectations in quality, quantity, and impact of contributions.
- Provided critical insights that shaped project planning and team success.

Linda Nguyen

The Co-operators

Rating: Outstanding

“Joshwyn was great with driving all of his tasks to completion and self-managed the various tasks. Documentation was concise and well structured. His presentation and communication skills were praised. . . Overall, Joshwyn was great to work with and it makes me very open to considering hiring another student in the future if they're anywhere near Joshwyn's dedication.”

- Independently drove tasks to completion with minimal oversight.
- Delivered high-quality automations, documentation, and presentations.
- Excellent adaptability, feedback incorporation, and collaboration.

Maureen Barrow

Dematic Logistics Systems

Rating: Outstanding

“Joshwyn is a highly valued member of our team, and his contributions have not gone unnoticed. His exceptional work ethic and positive attitude have made him a pleasure to work with. . . Overall, Joshwyn's contributions to our team have been invaluable, and we would be fortunate to have him return as a full-time member. Joshwyn is a terrific employee. I would rehire him in a heartbeat.”

- Highly valued team member with exceptional work ethic and professionalism.
- Strong presentation and communication skills; repeatedly commended by colleagues.

All testimonials provided by direct supervisors during end of term co-op performance evaluations.